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Crop protection :: IPM :: Cotton

IPM-Cotton

Monitoring for Pest & Disease

- Undertake community level monitoring to know change in distribution and abundance of pest.
- Organize regular pest monitoring and assess bio control potential at every 10 days interval.
- Record pest incidence at every 5 to 10 km distance at 12 spots/ha and 5 plants/spot selected randomly.
- Use pheromone traps for monitoring of American, spotted & pink bollworm and spodoptera, at a distance of 50 m @ 5 traps/ha.
- Set up yellow pan/sticky traps for monitoring whitefly @ 25 yellow pans/sticky traps/ha.
- Monitor the incidence of sucking pests and apply botanical pesticides if the seed treatment fails to offer desired level of protection.
- Setup pheromone traps (5/ha) at the distance of 50 meters to lure the adult moths for monitoring.
- Based on the incidence/observation of the traps, a second release of the egg parasitoids and predators should be done.

Pre-sowing stage

- Deep ploughing in summer
- Removal of alternate hosts.
- Avoid cotton after cotton.
- Adopt crop rotation.
- Remove and destroy crop residues by burning
- Acid delinting of seeds should be done before sowing (with commercial sulphuric acid @ 0.1 l/kg of seed) and fumigate with celphos @ 3g/cum for 48 hrs and get thoroughly air dried.

Sowing date

- Grow tolerant/resistant varieties of key pests of that area
- Use certified seeds
- Acid delinting treatment for seeds
- Seed dipping in antibiotic in black arm endemic areas
- Early sowing
- Seed treatment with insecticides
- Use pre-emergence/ post emergence herbicides
- Adopt proper spacing and maintain plant population.
- Adopt judicious fertilizer and water management. Avoid excessive use of nitrogenous fertilizers. Add FYM @ 25 tonnes/ha.
- Maintain weed-free crop for at least 8-9 weeks after sowing till canopy starts closing by timely inter- cultivation.
- Trap cropping should be done with crops like okra, castor, marigold, *Nicotiana rustica*, and insect feeding on these, should be periodically removed or killed.
- Inter-cropping with mung bean, soybean, groundnut, ragi, maize, cowpea and onion reduces the infestation of boll worms. However, wild brinjal reduces the whitefly incidence. Intercropping with cowpea reduces Fusarium wilt and colonise the bio control fauna such as lady bird beetles, Chrysoperla etc.

Vegetative stage

- Gap filling and thinning
- Inter culture & hand weeding of weeds
- Check population on trap crops & inter crops
- Release of Chrysoperla grubs @ 1000/ha.
- Spray neem products for whitefly
- If pest persists spray recommended insecticides
- Crushing of larvae in the shoots mechanically
- Setup pheromone traps (5/ha) at the distance of 50 meters to lure the adult moths for monitoring.
- Fix yellow sticky traps for white fly
- Remove & destroy root rot affected plants

Flowering stage

- Continue inter culturing & hand weeding of weeds
- Management of trap crops & inter crops
- Release Chrysoperla @ 10,000/ha for sucking pest
- Use pheromone traps and change lures for bollworms
- Management of population of bollworms in trap crops
- Release of Trichogramma @ 1.5 lac/ha for bollworms
- Set up bird perches
- Use yellow sticky traps for whitefly
- Use neem product to control whitefly
- If not controlled, spray triazophos/acephae/acetamprid
- Destroy plants affected with cotton leaf curly virus
- Spray recommended chemical for vector control

Fruiting stage

- Collection & destruction of damaged floral bodies
- Collection of grown up larvae under destruction.
- Use Ha.NPV @ 250-500 LE/ha
- Use neem products
- Removal of terminals is to be done
- Spray Spodoptera NPV in evening hours.
- Spray recommended insecticides

Boll opening stage

- Apply need based recommended insecticides
- Do not extend the crop period
- Use monitoring device
- Use recommended acaricides for mites
- Destruction of CLCV infected plants
- Spot application of chemicals

Storage

- Terminate the crop at maturity as early as possible and avoid ratooning to prevent the re- infestation of insect pests/diseases.
- Allow grazing by animals after the last picking for checking the carry over population of bollworms (animals feed on plant debris and unopened bolls).
- Dispose gin trashes by burning and ginning the seeds in the rollers to eliminate carry over of pink bollworm. Install PBW traps in ginneries.
- Avoid staking of the cotton stalks near the fields. Destroy the opened bolls if any on the plant before staking.
- Crushing of cotton seeds to be completed by April end.
- Fumigation of seeds may be undertaken with expert supervision.